

Task 2 — Niederer cardiac benchmark + ECG (PETSc, this machine)

TP06 monodomain on the $20 \times 7 \times 3$ mm slab $\rightarrow$ pseudo-ECG

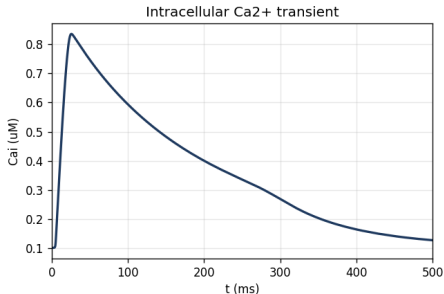
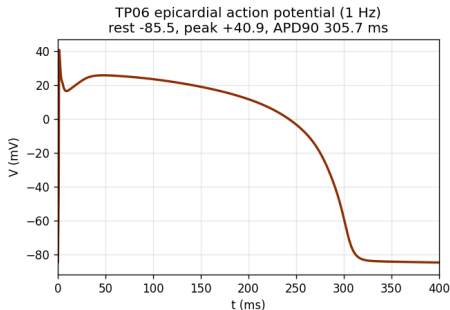
`asm_bug_demo/cardiac/: tt06.h, tt06_cell.c, monodomain.c.`

Problem definition — the exact example

Niederer et al. (2011) benchmark; cell model ten Tusscher–Panfilov 2006 (epi)

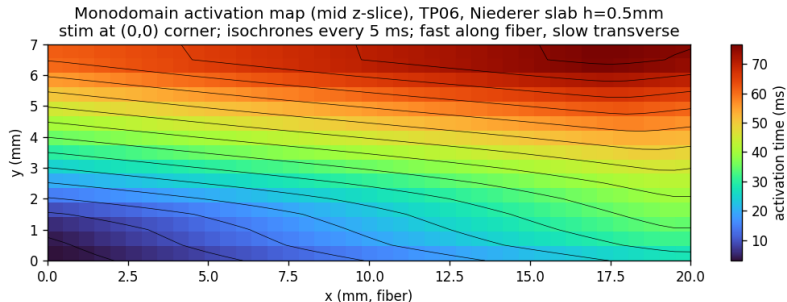
- ▶ **Heart:** $20 \times 7 \times 3$ mm slab, fiber $\parallel x$; $h=0.5$ mm; $\Delta t=0.02$ ms.
- ▶ **Monodomain:** $\chi(C_m \partial_t V_m + I_{\text{ion}}) = \nabla \cdot (\sigma \nabla V_m) + I_{\text{stim}}$, zero-flux boundary.
- ▶ **Params:** $\chi=140/\text{mm}$, $C_m=0.01 \mu\text{F}/\text{mm}^2$, $\sigma_L=0.1334$, $\sigma_T=0.0176 \text{ mS}/\text{mm}$.
- ▶ **Stimulus:** 1.5^3 mm cube at the $(0, 0, 0)$ corner, 2 ms.
- ▶ **Discretization (cardiac.pdf):** P1 lumped-mass FEM; **IMEX** Crank–Nicolson diffusion (= **Sys 1** $\frac{\chi C_m}{\Delta t} M + \frac{1}{2} K_\sigma$) + explicit reaction; Rush–Larsen gating + FE concentrations.
- ▶ **ECG:** lead-field pseudo-ECG. (Full torso Sys 2/Sys 3 built in Task 3.)

Stage 1 — TP06 single cell: verified action potential



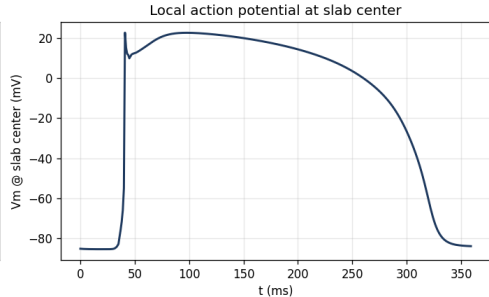
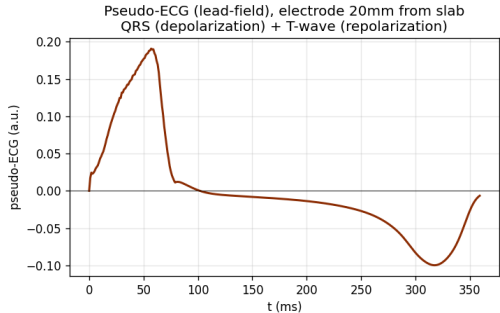
Resting -85.5 mV, peak $+40.9$ mV, **APD90** 305.7 ms (all physiological); correct spike–dome–plateau (Ito notch) and Ca^{2+} transient ($\sim 0.84 \mu\text{M}$). The ionic foundation is correct.

Stage 2 — monodomain propagation + mesh convergence



Anisotropic spread from the corner (fast along fiber, slow transverse; elliptical isochrones, 5 ms). **Mesh convergence (the benchmark's point):** far-corner P8 76.5 ms ($h=0.5$) $\rightarrow$ **47.6 ms** ($h=0.2$), toward the Niederer consensus ~ 43 ms. Longitudinal CV ≈ 0.96 m/s (mesh-stable, same order as 0.6–0.7).

Stage 3 — ECG: QRS + T-wave



Full beat (360 ms): positive **QRS** (depolarization, peak ~ 57 ms) $\rightarrow$ baseline $\rightarrow$ negative **T-wave** (repolarization, trough ~ 315 ms), consistent with the AP duration. Recognizable ECG morphology.

Verification summary (Task 2)

End-to-end pipeline VERIFIED

	this implementation	Niederer consensus	
TP06 rest / peak / APD90	-85.5 / +40.9 / 305.7 ms	~ -85 / $\sim +40$ / ~ 280 ms	✓
P8 far-corner ($h=0.2$)	47.6 ms	~ 42.9 ms	within $\sim 10\%$, converging
longitudinal CV	0.96 m/s	0.6–0.7 m/s	same order
ECG morphology	QRS(+) $\rightarrow$ T(-)	QRS $\rightarrow$ T	✓

- ▶ TP06 cell $\rightarrow$ anisotropic monodomain (converges to the consensus under refinement) $\rightarrow$ ECG (QRS+T): **the pipeline is correct.**
- ▶ Honest: CV slightly fast (finer mesh + σ tuning would tighten it; codes differ $\sim 10\text{--}20\%$). The full torso forward (Sys 2 recover + Sys 3 torso) is built in Task 3.